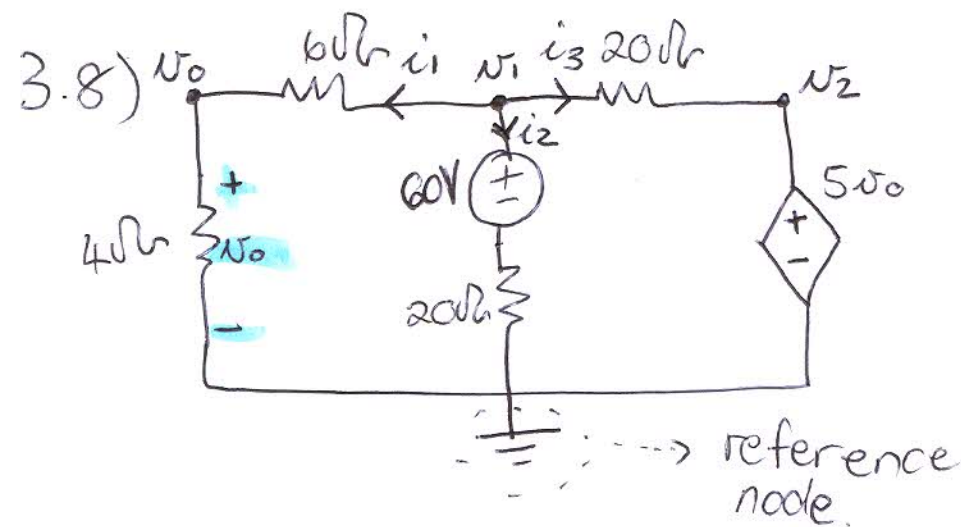


Tutorial 3

28 Feb '14



Find v_0 ?

Use nodal analysis.

But $v_2 = 5v_0$

KCL node v_1 : $i_1 + i_2 + i_3 = 0$

$$\frac{v_1 - v_0}{6} + \frac{v_1 - 60}{20} + \frac{v_1 - v_2}{20} = 0$$

$$20(v_1 - v_0) + 6(v_1 - 60) + 6(v_1 - 5v_0) = 0$$

$$32v_1 - 50v_0 = 360 \quad \dots (1)$$

using ohm's law.

$$v_0 = i_1 R = i_1 (4)$$

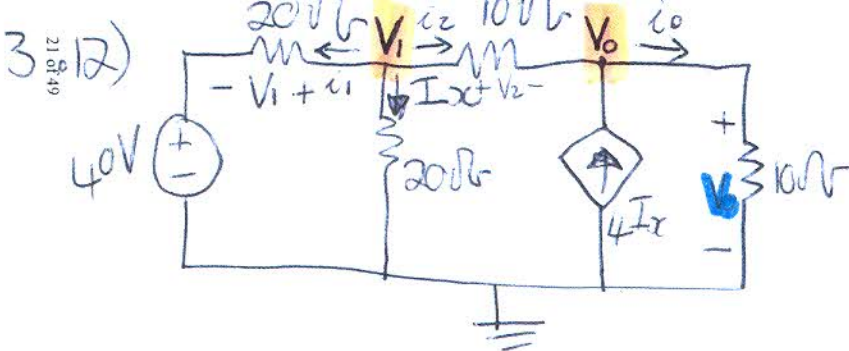
$$v_0 = \left(\frac{v_1 - v_0}{6} \right) (4)$$

$$\frac{5}{2}v_0 = v_1 \quad \dots (2)$$

Substitute (2) in (1)

$$32\left(\frac{5}{2}v_0\right) - 50v_0 = 360$$

$$\underline{v_0 = 12 \text{ V}}$$

Find V_0 ?

Use Nodal analysis.

2 unknown nodes: V_1 & V_0 KCL Node 1: $i_1 + I_x + i_2 = 0$

$$\frac{V_1 - 40}{20} + \frac{V_1 - 0}{20} + \frac{V_1 - V_0}{10} = 0$$

x(10)x(20)

$$10V_1 - 400 + 10V_1 + 20V_1 - 20V_0 = 0$$

$$40V_1 - 20V_0 = 400 \quad \text{--- (1)}$$

KCL Node 2: $i_2 + 4I_x - i_0 = 0$

$$\frac{V_1 - V_0}{10} + 4\left(\frac{V_1 - 0}{20}\right) - \frac{V_0 - 0}{10} = 0$$

$$20V_1 - 20V_0 + 40V_1 - 20V_0 = 0$$

$$60V_1 - 40V_0 = 0$$

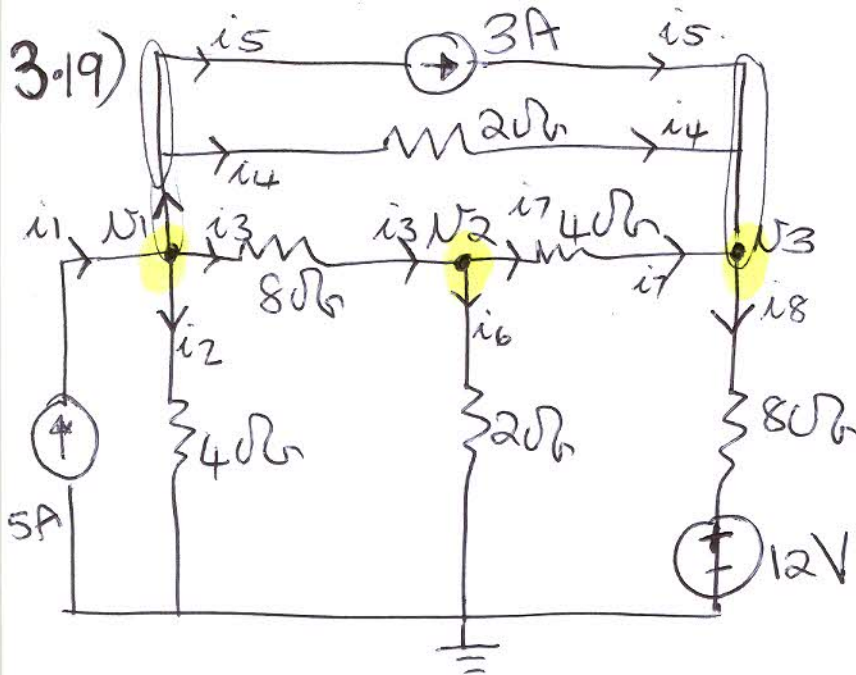
$$60V_1 = 40V_0$$

$$V_1 = \frac{2}{3}V_0 \quad \text{--- (2)}$$

vervang (2) in (1)

$$40\left(\frac{2}{3}V_0\right) - 20V_0 = 400$$

$$V_0 = 60V \rightarrow$$



Use nodal analysis

→ Find v_1, v_2 & v_3

$$i_1 = 5A$$

$$i_5 = 3A$$

KCL at node 1: $i_1 = i_2 + i_3 + i_4 + i_5$

$$5 = \frac{v_1 - 0}{4} + \frac{v_1 - v_2}{8} + \frac{v_1 - v_3}{2} + 3$$

$$8(2) = 2v_1 + v_1 - v_2 + 4v_1 - 4v_3$$

$$7v_1 - v_2 - 4v_3 = 16 \quad \text{--- (1)}$$

KCL at node 2: $i_3 = i_6 + i_7$

$$\frac{v_1 - v_2}{8} = \frac{v_2 - 0}{2} + \frac{v_2 - v_3}{4}$$

$$v_1 - v_2 = 4v_2 + 2v_2 - 2v_3$$

$$v_1 - 7v_2 + 2v_3 = 0 \quad \text{--- (2)}$$

KCL at node 3: $i_5 + i_4 + i_7 = i_8$

$$3 + \frac{v_1 - v_3}{2} + \frac{v_2 - v_3}{4} = \frac{v_3 - 12}{8}$$

$$24 + 4v_1 - 4v_3 + 2v_2 - 2v_3 = v_3 - 12$$

$$4v_1 + 2v_2 - 7v_3 = -36 \quad \text{--- (3)}$$

Write (1), (2), (3) in matrix form

$$\begin{bmatrix} 7 & -1 & -4 \\ 1 & -7 & 2 \\ 4 & 2 & -7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ -36 \end{bmatrix}$$

$$[A][V] = [B]$$

For Δ ? Repeat 1st 2 rows and cross multiply.

$$\Delta = \begin{vmatrix} 7 & -1 & -4 \\ 1 & -7 & 2 \\ 4 & 2 & -7 \end{vmatrix} \begin{matrix} \rightarrow + \\ \rightarrow + \\ \rightarrow + \end{matrix}$$

$$= 343 + (-8) + (-8) - (112 + 28 + 7) = \underline{180}$$

For: $V_1 = \frac{\Delta_1}{\Delta}$ - ? Replace $[B]$ in desired column, repeat first 2 rows.

$$\Delta_1 = \begin{vmatrix} 16 & -1 & -4 \\ 0 & -7 & 2 \\ -36 & 2 & -7 \end{vmatrix} \begin{matrix} \rightarrow \oplus \\ \rightarrow \oplus \\ \rightarrow \oplus \end{matrix}$$

$$= 784 + 0 + 72 - (-1008 + 64 + 0) = \underline{1800}$$

$$V_1 = \frac{1800}{180} = 10$$

$$V_2 = \frac{888}{180} = 4.933$$

$$\Delta_2 = \begin{vmatrix} 7 & 16 & -4 \\ 1 & 0 & 2 \\ 4 & -36 & -7 \end{vmatrix} \begin{matrix} \rightarrow \oplus \\ \rightarrow \oplus \\ \rightarrow \oplus \end{matrix}$$

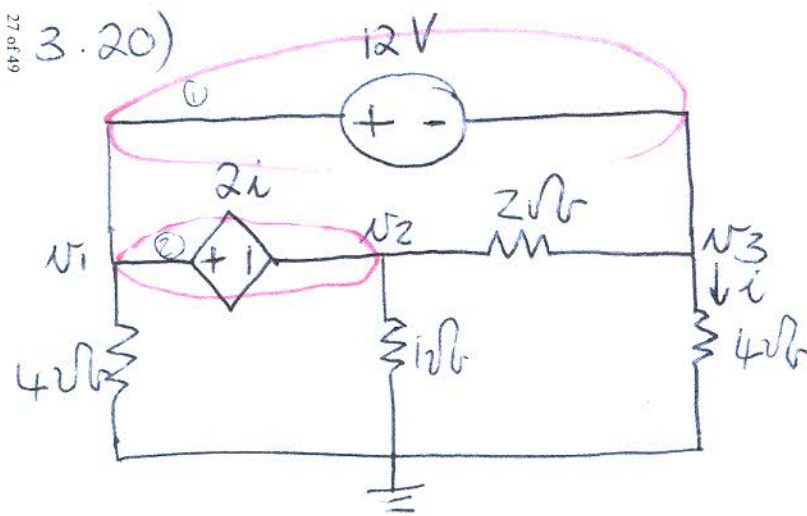
$$\Delta_3 = \begin{vmatrix} 7 & -1 & 16 \\ 1 & -7 & 0 \\ 4 & 2 & -36 \end{vmatrix} \begin{matrix} \rightarrow \oplus \\ \rightarrow \oplus \\ \rightarrow \oplus \end{matrix}$$

$$V_3 = \frac{2208}{180} = 12.267$$

$$V = \begin{bmatrix} 10 \\ 4.933 \\ 12.267 \end{bmatrix}$$

$$= 0 + 144 + 128 - (0 - 504 - 112) = 1764 + 32 + 0 - (-448 + 0 + 36) = 2208$$

(4)

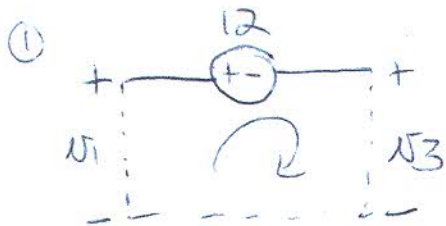


Find: v_1, v_2, v_3

Use: Nodal analysis.

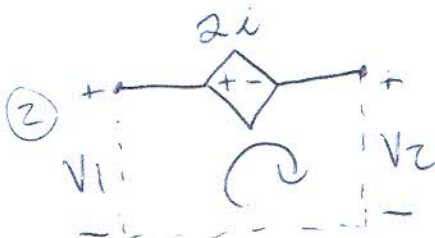
$$i = \frac{v_3 - 0}{4}$$

Two super nodes.



$$\text{KVL: } -v_1 + 12 + v_3 = 0$$

$$v_3 = v_1 - 12 \quad \text{--- ①}$$



$$\text{KVL: } -v_1 + 2i + v_2 = 0$$

$$v_1 = 2i + v_2 \quad \text{---}$$

Substitute i

$$v_1 = \frac{2v_3}{4} + v_2$$

$$v_1 = 0.5v_3 + v_2 \quad \text{--- ②}$$

Substitute ① in ②

$$v_1 = 0.5(v_1 - 12) + v_2$$

$$+0.5v_1 = -6 + v_2$$

$$v_1 = -12 + 2v_2 \quad \text{--- ③}$$

Substitute ③ in ④

$$-12 + 2v_2 + 4v_2 + v_3 = 0$$

$$6v_2 + (v_1 - 12) - 12 = 0$$

$$6v_2 + (-12 + 2v_2) - 24 = 0$$

$$8v_2 = 36$$

$$v_2 = 4.5 \text{ V}$$

From the supernode:

$$\frac{v_1}{4} + \frac{v_2}{1} + \frac{v_3}{4} = 0$$

$$v_1 + 4v_2 + v_3 = 0 \quad \text{--- ④}$$

$$\text{Then } v_1 = -12 + 2(4.5)$$

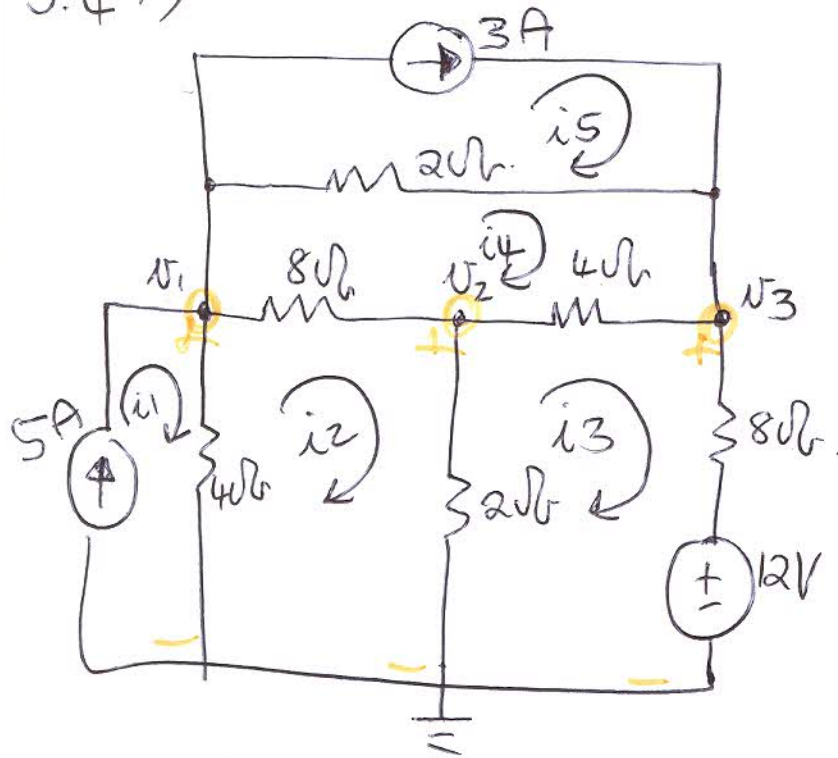
$$= -3 \text{ V}$$

$$v_3 = -3 - 12$$

$$= -15 \text{ V}$$

⑤

3.47)

use mesh analysis→ Find v_1, v_2, v_3 ?

From mesh currents:

$$i_1 = 5 \text{ A}$$

$$i_5 = 3 \text{ A}$$

KVL mesh 2: $4(i_2 - i_1) + 8(i_2 - i_4) + 2(i_2 - i_3) = 0$
 $14i_2 - 2i_3 - 8i_4 = 20 \dots\dots ①$

KVL mesh i3: $12 + 2(i_3 - i_2) + 4(i_3 - i_4) + 8i_3 = 0$
 $-2i_2 + 14i_3 - 4i_4 = -12 \dots\dots ②$

KVL mesh i4: $8(i_4 - i_2) + 2(i_4 - i_5) + 4(i_4 - i_3) = 0$
 $-8i_2 - 4i_3 + 14i_4 = 6 \dots\dots ③$

$$\begin{bmatrix} 14 & -2 & -8 \\ -2 & 14 & -4 \\ -8 & -4 & 14 \end{bmatrix} \begin{bmatrix} i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 20 \\ -12 \\ 6 \end{bmatrix} \therefore \begin{aligned} i_2 &= 2.5 \text{ A} \\ i_3 &= 0.0333 \text{ A} \\ i_4 &= 1.8667 \text{ A} \end{aligned}$$

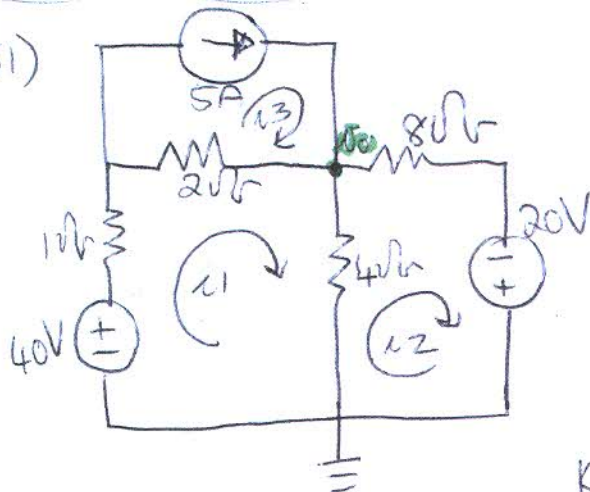
$$v_1 = iR = (i_1 - i_2)(4) = (5 - 2.5)(4) = 10 \text{ V}$$

$$v_2 = iR = (i_2 - i_3)(2) = (2.5 - 0.0333)(2) = 4.9334 \text{ V}$$

$$v_3 = i_3 R + 12 = (0.0333)(8) + 12 = 12.2664 \text{ V}$$

(6)

3.51)

Find v_o ?

$$\text{KVL mesh 1: } -40 + i_1 + 2(i_1 - i_3) + 4(i_1 - i_2) = 0$$

$$7i_1 - 4i_2 - 2i_3 = 40 \quad (1)$$

$$\text{KVL mesh 2: } 8i_2 - 20 + 4(i_2 - i_1) = 0$$

$$-4i_1 + 12i_2 = 20$$

$$-2i_1 + 6i_2 = 10 \quad (2)$$

$$\text{KVL mesh 3: } i_3 = 5A \quad (3)$$

$$\text{Substitute (3) in (1)} \quad 7i_1 - 4i_2 = 50 \quad (5)$$

$$\text{From (2): } -2i_1 = 10 - 6i_2$$

$$i_1 = -5 + 3i_2 \quad (4)$$

Substitute (4) in (5)

$$7(-5 + 3i_2) - 4i_2 = 50$$

$$17i_2 = 85$$

$$i_2 = 5A$$

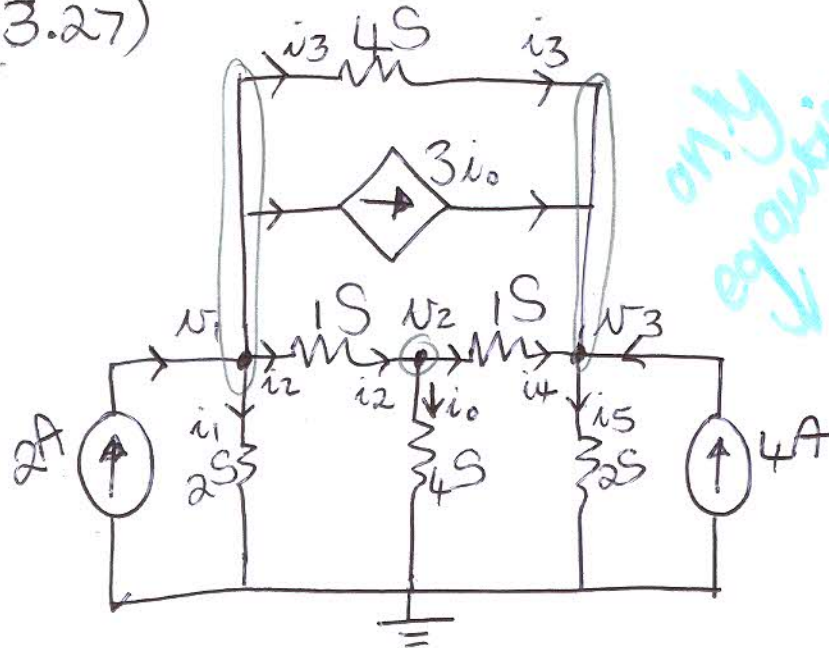
$$i_1 = -5 + 3(5)$$

$$= 10A$$

$$v_o = iR = (i_1 - i_2)(4)$$

$$= (10 - 5)(4) = 20V$$

3.27)



only equations

Find v_1, v_2, v_3

Use nodal analysis

$$G = \frac{1}{R}$$

$$R = \frac{1}{G}$$

KCL node v_1 : $2 = i_1 + i_2 + 3i_0 + i_3$

$$2 = \frac{v_1 - 0}{\frac{1}{2}} + \frac{v_1 - v_2}{\frac{1}{1}} + 3\left(\frac{v_2 - 0}{\frac{1}{4}}\right) + \frac{v_1 - v_3}{\frac{1}{4}}$$

$$2 = 2v_1 + v_1 - v_2 + 12v_2 + 4v_1 - 4v_3$$

$$7v_1 + 11v_2 - 4v_3 = 2 \quad \dots \textcircled{1}$$

KCL node v_2 : $i_2 = i_0 + i_4$

$$\frac{v_1 - v_2}{\frac{1}{1}} = \frac{v_2}{\frac{1}{4}} + \frac{v_2 - v_3}{\frac{1}{1}}$$

$$v_1 - v_2 = 4v_2 + v_2 - v_3$$

$$v_1 - 6v_2 + v_3 = 0 \quad \dots \textcircled{2}$$

KCL node v_3 : $i_4 + 4 + 3i_0 + i_3 = i_5$

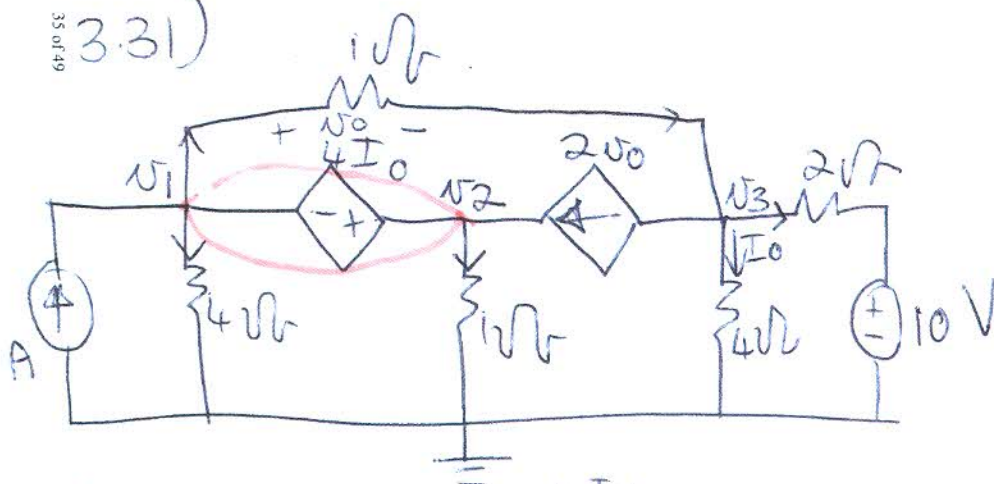
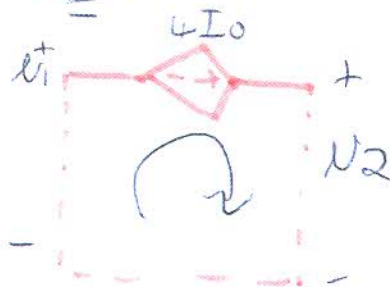
$$v_2 - v_3 + 4 + 3(4v_2) + 4(v_1 - v_3) = 2v_3$$

$$4v_1 + 13v_2 - 7v_3 = -4 \quad \dots \textcircled{3}$$

$$\begin{bmatrix} 7 & 11 & -4 \\ 1 & -6 & 1 \\ 4 & 13 & -7 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

⑧

3.31)

Find v_1, v_2, v_3 .Supernode:

$$\text{KVL: } -v_1 - 4I_0 + v_2 = 0$$

$$I_0 = \frac{v_3 - 0}{4}$$

$$\therefore -v_1 - v_3 + v_2 = 0 \quad \text{--- (1)}$$

Supernode KCL:

$$1 - \frac{v_1}{4} - \frac{v_1 - v_3}{1} - \frac{v_2}{1} + 2v_0 = 0$$

$$\text{and } v_0 = v_1 - v_3$$

$$4 - v_1 - 4v_1 + 4v_3 - 4v_2 + 8(v_1) - 8v_3 = 0$$

$$+ 3v_1 - 4v_2 - 4v_3 = -4 \quad \text{--- (2)}$$

KCL node 3:

$$2v_0 + I_0 + \frac{v_3 - 10}{2} - \left(\frac{v_1 - v_3}{1} \right) = 0$$

$$4v_0 + 2I_0 + v_3 - 10 - 2v_1 + 2v_3 = 0$$

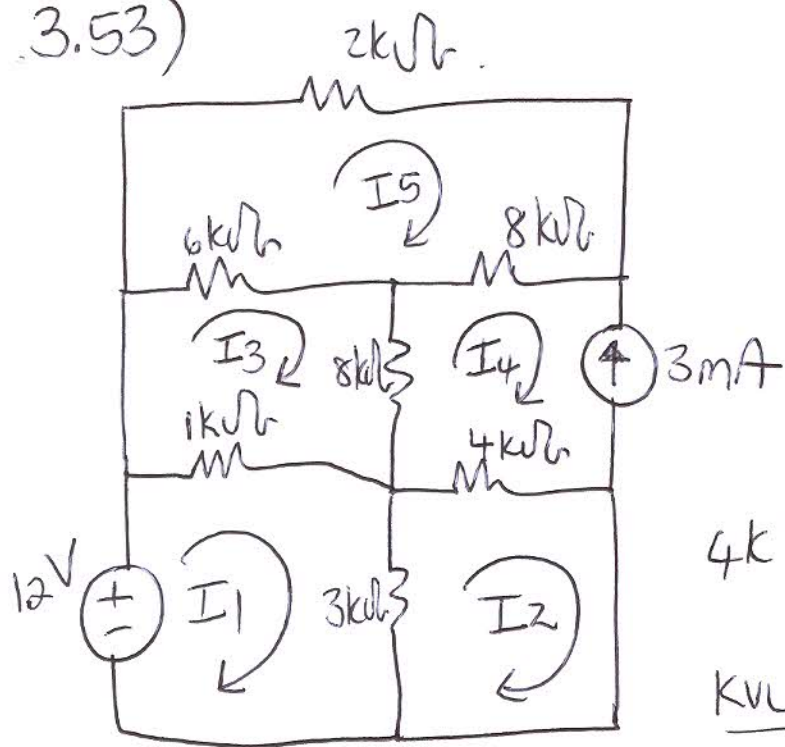
$$4v_1 - 4v_3 + 0.5v_3 + 3v_3 - 2v_1 = 10$$

$$2v_1 - 0.5v_3 = 10$$

$$4v_1 - 1v_3 = 20$$

$$\begin{bmatrix} -1 & 1 & -1 \\ 3 & -4 & -4 \\ 4 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 20 \end{bmatrix}$$

3.53)



Find mesh current equations.

KVL mesh I1:

$$-12 + 1k(I_1 - I_3) + 3k(I_1 - I_2) = 0$$

$$4kI_1 - 3kI_2 - 1kI_3 + 0I_4 + 0I_5 = 12 \quad \dots (1)$$

KVL mesh I2:

$$3k(I_2 - I_1) + 4k(I_2 - I_4) = 0 \quad \dots (2)$$

$$-3kI_1 + 7kI_2 + 0I_3 - 4kI_4 + 0I_5 = 0$$

$$1k(I_3 - I_1) + 8k(I_3 - I_4) + 6k(I_3 - I_5) = 0$$

$$-1kI_1 + 0I_2 + 15kI_3 - 8kI_4 - 6kI_5 = 0 \quad \dots (3)$$

KVL mesh I4: $I_4 = -3mA \quad \dots (4)$

KVL mesh I5:

$$6k(I_5 - I_3) + 8k(I_5 - I_4) + 2kI_5 = 0$$

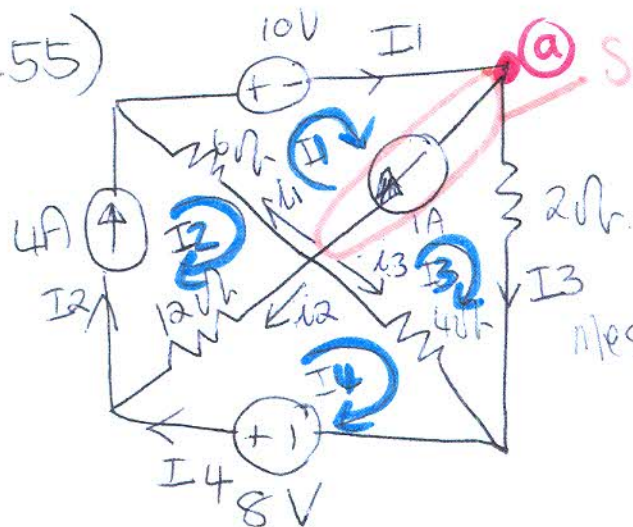
$$-6kI_3 - 8kI_4 + 16kI_5 = 0 \quad \dots (5)$$

$$\begin{bmatrix} 4k & -3k & -1k & 0 & 0 \\ -3k & 7k & 0 & -4k & 0 \\ -1k & 0 & 15k & -8k & -6k \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -6k & -8k & 16k \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \\ -3mA \\ 0 \end{bmatrix}$$

↑ you can also substitute I_4 which gives us a 4 row matrix

(10)

3.55)



Supernode!

Find i_1, i_2, i_3 ?

$$\text{Mesh 2: } 6(I_2 - I_1) + 12(I_2 - I_4) = 0$$

$$-6I_1 + 18I_2 - 12I_4 = 0$$

$$-3I_1 + 9I_2 - 6I_4 = 0 \quad \text{--- (1)}$$

$$\text{Mesh 4: } -8 + 12(I_4 - I_2) + 4(I_4 - I_3) = 0$$

$$12I_2 - 4I_3 + 16I_4 = 8$$

$$3I_2 - I_3 + 4I_4 = 2 \quad \text{--- (2)}$$

$$\text{Supermesh: } 10 + 2I_3 + 4(I_3 - I_4) + 6(I_1 - I_2) = 0$$

$$6I_1 - 6I_2 + 6I_3 - 4I_4 = -10$$

$$3I_1 - 3I_2 + 3I_3 - 2I_4 = -5 \quad \text{--- (3)}$$

$$I_2 = 4A \rightarrow \text{Substitute in eqn (1) - (3)}$$

KCL a:

$$I_1 = 1 + I_3 \quad \text{---}$$

$$\begin{bmatrix} -3 & 9 & 0 & -6 \\ 0 & 3 & -1 & 4 \\ 3 & -3 & 3 & -2 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -5 \\ 1 \end{bmatrix}$$

Calculate $I_1 - I_4$

$$\text{Now: } i_1 = I_1 - I_2$$

$$i_2 = I_2 - I_4$$

$$i_3 = I_4 - I_3$$

